

3a – algorithmic graphs

GR

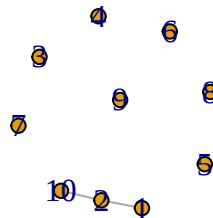
2024-08-20

Load libraries

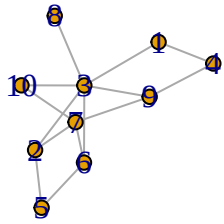
```
library(igraph)
library(tidyverse)
library(ggnetwork)
```

Erdos-Renyi (“random”)

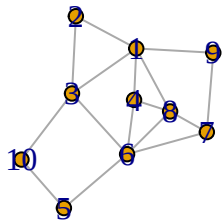
```
sample_gnp(n=10,p=.1) %>% plot.igraph()
```



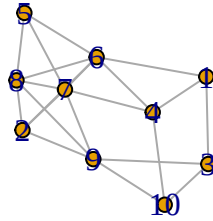
```
sample_gnp(n=10,p=.2) %>% plot.igraph()
```



```
sample_gnp(n=10,p=.3) %>% plot.igraph()
```

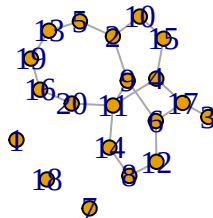


```
sample_gnp(n=10,p=.4) %>% plot.igraph()
```



<https://www.netlogoweb.org/launch#https://www.netlogoweb.org/assets/modelslib/Sample%20Models/Networks/Giant%20Component.nlogo>

```
er_20_20.g <- sample_gnm(n=20,m=20)
plot.igraph(er_20_20.g)
```



```
ecount(er_20_20.g)
```

```
## [1] 20
```

```
vcount(er_20_20.g)
```

```
## [1] 20
```

```
components(er_20_20.g)$no
```

```
## [1] 4
```

```
diameter(er_20_20.g)
```

```
## [1] 6
```

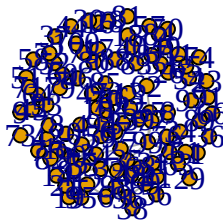
```
mean_distance(er_20_20.g)
```

```
## [1] 2.985294
```

```
mean(harmonic_centrality(er_20_20.g))
```

```
## [1] 5.888333
```

```
er_100_100.g <- sample_gnm(n=100,m=100)  
plot.igraph(er_100_100.g)
```



```
ecount(er_100_100.g)
```

```
## [1] 100
```

```
vcount(er_100_100.g)
```

```
## [1] 100
```

```
components(er_100_100.g)$no
```

```
## [1] 15
```

```
diameter(er_100_100.g)
```

```
## [1] 13
```

```
mean_distance(er_100_100.g)
```

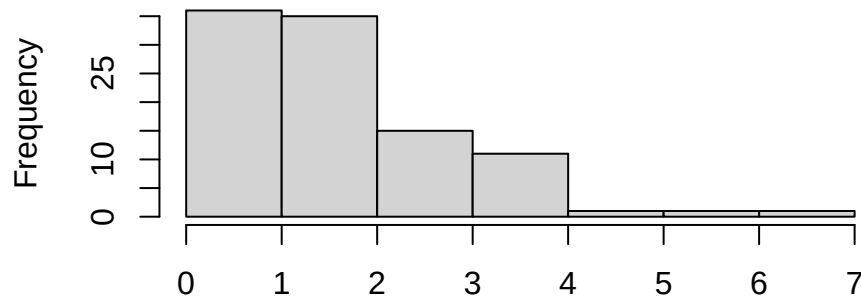
```
## [1] 5.918315
```

```
mean(harmonic_centralty(er_100_100.g))
```

```
## [1] 15.43243
```

```
er_100_100.g %>% degree() %>% hist()
```

Histogram of .



Poisson with mean of 2.

```
# you really don't need to know all this stuff with for loops and matrices  
# unless you want to do a senior thesis using simulations or the like  
trials <- 20  
testvalues <- seq(5,250,5)  
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)  
colnames(results.mat) <- c("size", "apl", "mhc", "diameter")  
j <- 1  
for (i in testvalues) {  
  for (k in 1:trials) {  
    g <- sample_gnm(n=i,m=i)  
    apl <- mean_distance(g)
```

```

mhc <- harmonic_centrality(g) %>% mean()
diameter <- diameter(g)
results.mat[j,] <- c(i, apl, mhc, diameter)
j <- j+1
}
}

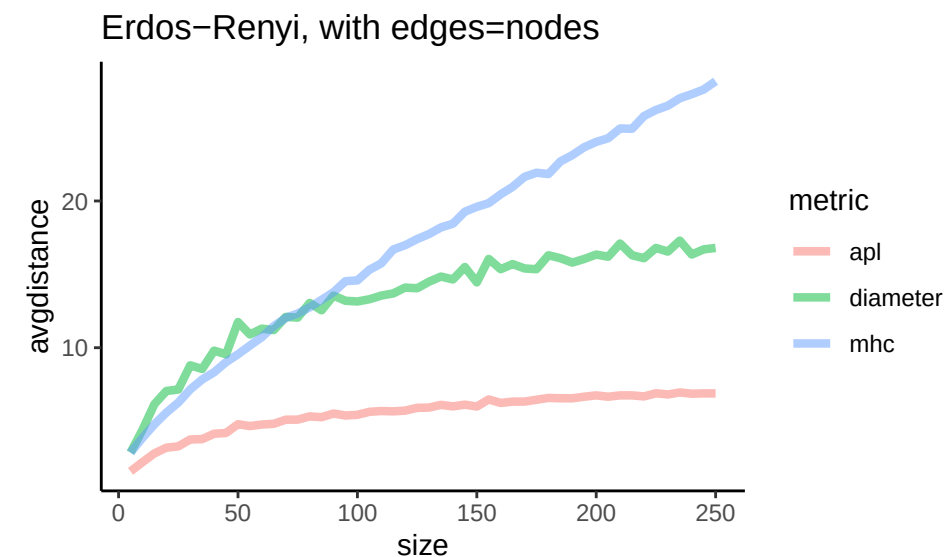
results.mat %>%
  as_data_frame() %>%
  group_by(size) %>%
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Erdos-Renyi, with edges=nodes")

```

```

## Warning: 'as_data_frame()' was deprecated in tibble 2.0.0.
## i Please use 'as_tibble()' (with slightly different semantics) to convert to a
## tibble, or 'as.data.frame()' to convert to a data frame.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

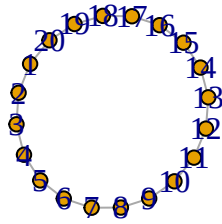


Ring

```

ring20.g <- make_ring(20)
plot.igraph(ring20.g)

```



```
ecount(ring20.g)
```

```
## [1] 20
```

```
vcount(ring20.g)
```

```
## [1] 20
```

```
components(ring20.g)$no
```

```
## [1] 1
```

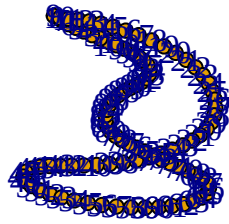
```
mean_distance(ring20.g)
```

```
## [1] 5.263158
```

```
mean(harmonic_centrality(ring20.g))
```

```
## [1] 5.757937
```

```
ring100.g <- make_ring(100)  
plot.igraph(ring100.g)
```



```
ecount(ring100.g)
```

```
## [1] 100
```

```
vcount(ring100.g)
```

```
## [1] 100
```

```
components(ring100.g)$no
```

```
## [1] 1
```

```
mean_distance(ring100.g)
```

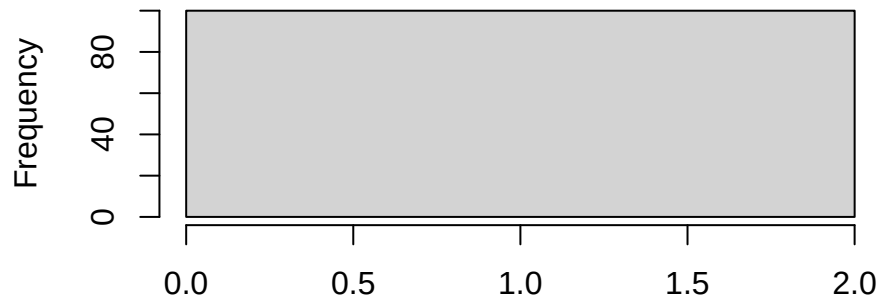
```
## [1] 25.25253
```

```
mean(harmonic_centrality(ring100.g))
```

```
## [1] 8.978411
```

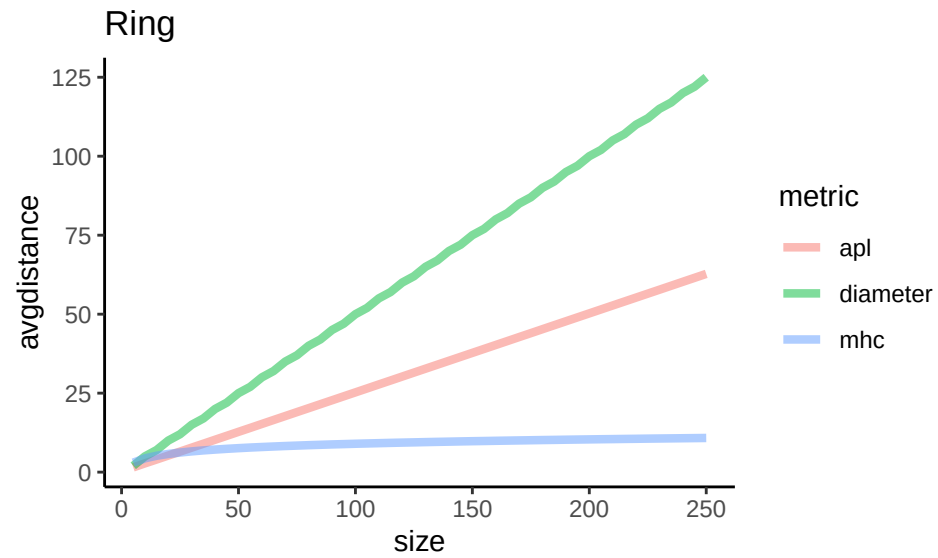
```
ring100.g %>% degree() %>% hist()
```

Histogram of .



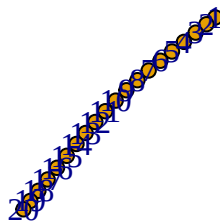
```
# you really don't need to know all this stuff with for loops and matrices
# unless you want to do a senior thesis using simulations or the like
trials <- 20
testvalues <- seq(5,250,5)
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)
colnames(results.mat) <- c("size", "apl", "mhc", "diameter")
j <- 1
for (i in testvalues) {
  for (k in 1:trials) {
    g <- make_ring(i)
    apl <- mean_distance(g)
    mhc <- harmonic_centralities(g) %>% mean()
    diameter <- diameter(g)
    results.mat[j,] <- c(i, apl, mhc, diameter)
    j <- j+1
  }
}

results.mat %>%
  as_data_frame() %>%
  group_by(size) %>%
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Ring")
```



Chain

```
chain20.g <- make_ring(20) %>% delete_edges("1|20")
plot.igraph(chain20.g)
```



```
ecount(chain20.g)
```

```
## [1] 19
```

```
vcount(chain20.g)
```

```
## [1] 20
```

```
components(chain20.g)$no
```

```
## [1] 1
```

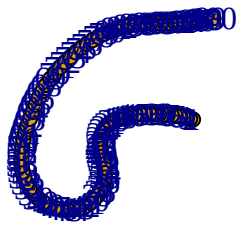
```
mean_distance(chain20.g)
```

```
## [1] 7
```

```
mean(harmonic_centrality(chain20.g))
```

```
## [1] 5.195479
```

```
chain100.g <- make_ring(100) %>% delete_edges("1|100")  
plot.igraph(chain100.g)
```



```
ecount(chain100.g)
```

```
## [1] 99
```

```
vcount(chain100.g)
```

```
## [1] 100
```

```
components(chain100.g)$no
```

```
## [1] 1
```

```
mean_distance(chain100.g)
```

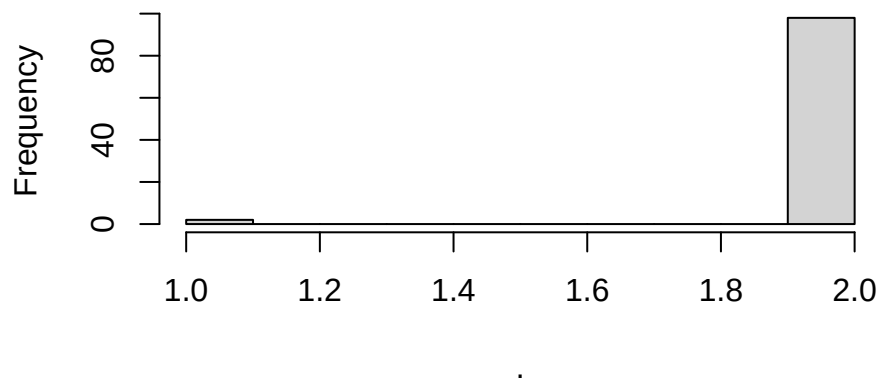
```
## [1] 33.66667
```

```
mean(harmonic_centrality(chain100.g))
```

```
## [1] 8.374755
```

```
chain100.g %>% degree() %>% hist()
```

Histogram of .

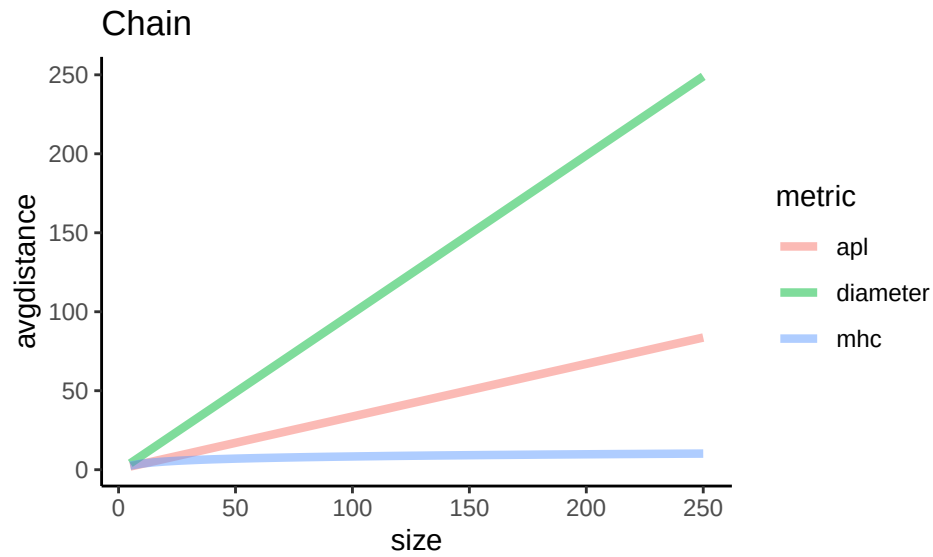


```
# you really don't need to know all this stuff with for loops and matrices  
# unless you want to do a senior thesis using simulations or the like  
trials <- 20  
testvalues <- seq(5,250,5)  
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)  
colnames(results.mat) <- c("size", "apl", "mhc", "diameter")  
j <- 1  
for (i in testvalues) {  
  for (k in 1:trials) {  
    g <- make_ring(i) %>% delete_edges("1|2")  
    apl <- mean_distance(g)  
    mhc <- harmonic_centrality(g) %>% mean()  
    diameter <- diameter(g)  
    results.mat[j,] <- c(i, apl, mhc, diameter)  
    j <- j+1  
  }  
}  
  
results.mat %>%  
  as_data_frame() %>%  
  group_by(size) %>%  
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
```

```

pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Chain")

```

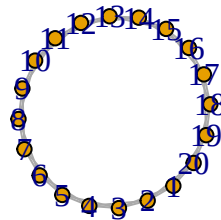


Watts-Strogatz (“small world”)

```

ws20_0.g <- sample_smallworld(dim=1, size=20, nei = 2, p=0)
plot.igraph(ws20_0.g)

```



```
ecount(ws20_0.g)
```

```
## [1] 40
```

```
vcount(ws20_0.g)
```

```
## [1] 20
```

```
components(ws20_0.g)$no
```

```
## [1] 1
```

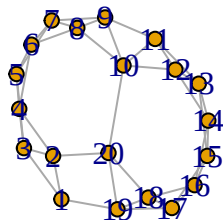
```
mean_distance(ws20_0.g)
```

```
## [1] 2.894737
```

```
mean(harmonic_centrality(ws20_0.g))
```

```
## [1] 8.933333
```

```
ws20_05.g <- sample_smallworld(dim=1, size=20, nei = 2, p=0.05)  
plot.igraph(ws20_05.g)
```



```
ecount(ws20_05.g)
```

```
## [1] 40
```

```
vcount(ws20_05.g)
```

```
## [1] 20
```

```
components(ws20_05.g)$no
```

```
## [1] 1
```

```
mean_distance(ws20_05.g)
```

```
## [1] 2.642105
```

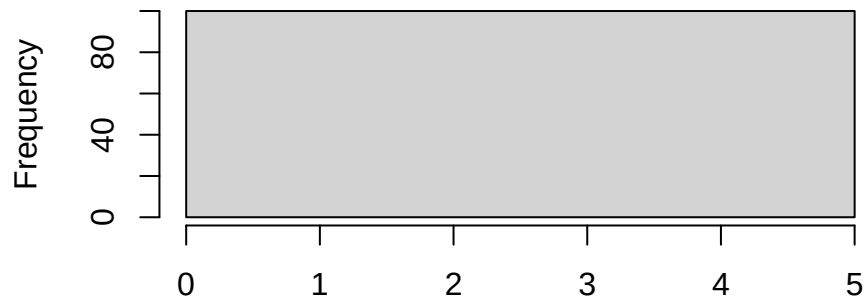
```
mean(harmonic_centrality(ws20_05.g))
```

```
## [1] 9.343333
```

<https://www.netlogoweb.org/launch#https://www.netlogoweb.org/assets/modelslib/Sample%20Models/Networks/Small%20Worlds.nlogo>

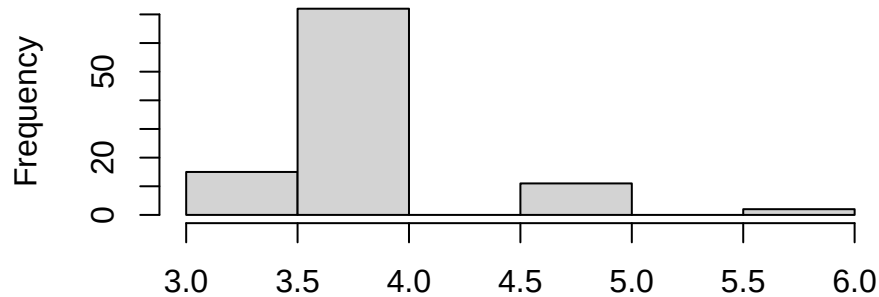
```
sample_smallworld(1,100,2,0) %>% degree() %>% hist()
```

Histogram of .



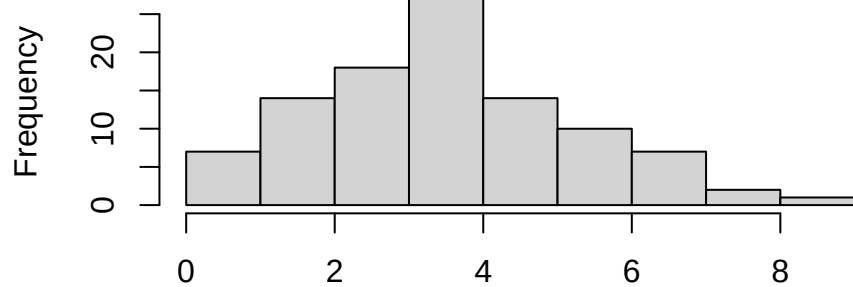
```
sample_smallworld(1,100,2,.05) %>% degree() %>% hist()
```

Histogram of .



```
sample_smallworld(1,100,2,1) %>% degree() %>% hist()
```

Histogram of .



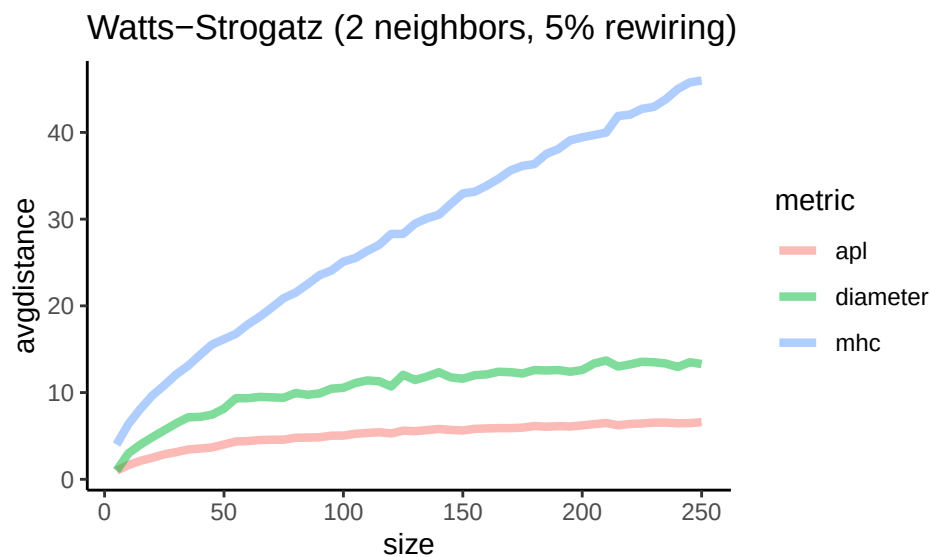
```
# you really don't need to know all this stuff with for loops and matrices
# unless you want to do a senior thesis using simulations or the like
trials <- 20
testvalues <- seq(5,250,5)
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)
colnames(results.mat) <- c("size", "apl", "mhc", "diameter")
j <- 1
for (i in testvalues) {
  for (k in 1:trials) {
    g <- sample_smallworld(dim=1, size=i, nei = 2, p=0.05)
    apl <- mean_distance(g)
    mhc <- harmonic_centrality(g) %>% mean()
    diameter <- diameter(g)
```

```

    results.mat[j,] <- c(i, apl, mhc, diameter)
    j <- j+1
  }
}

results.mat %>%
  as_data_frame() %>%
  group_by(size) %>%
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Watts-Strogatz (2 neighbors, 5% rewiring)")

```



```

# you really don't need to know all this stuff with for loops and matrices
# unless you want to do a senior thesis using simulations or the like
trials <- 100
testvalues <- seq(0,1,.01)
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 6)
colnames(results.mat) <- c("prop_rewire", "apl", "mhc", "diameter", "p_gc", "CC")
j <- 1
for (i in testvalues) {
  for (k in 1:trials) {
    g <- sample_smallworld(dim=1, size=100, nei = 2, p=i)
    apl <- mean_distance(g)
    mhc <- harmonic_centrality(g) %>% mean()
    diameter <- diameter(g)
    p_gc <- components(g)$csize %>% sort(decreasing = T)
    p_gc <- p_gc[1] / vcount(g)
    cc <- transitivity(g)
    results.mat[j,] <- c(i, apl, mhc, diameter, p_gc, cc)
    j <- j+1
  }
}

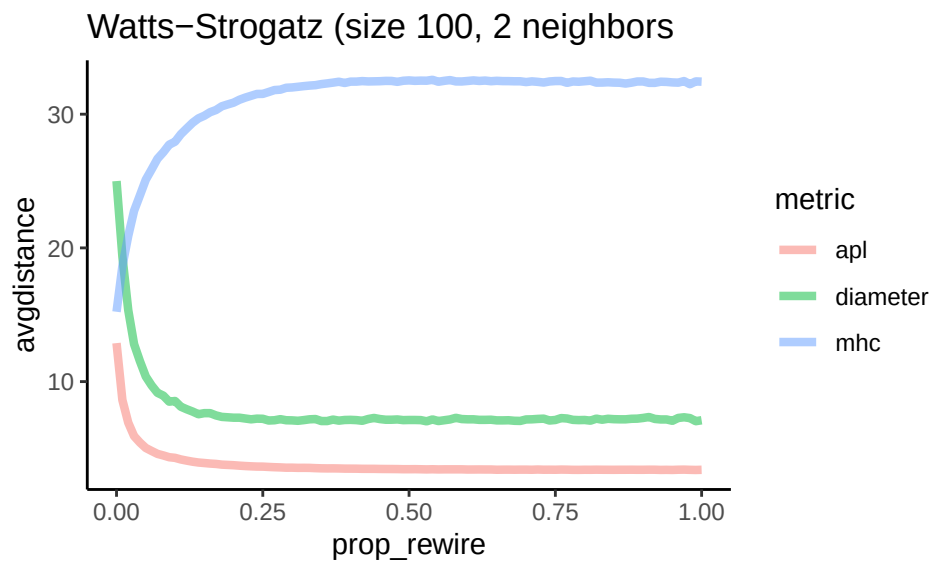
```

```

}

results.mat %>%
  as_data_frame() %>%
  group_by(prop_rewire) %>%
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
  pivot_longer(!prop_rewire, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=prop_rewire, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Watts-Strogatz (size 100, 2 neighbors)")

```

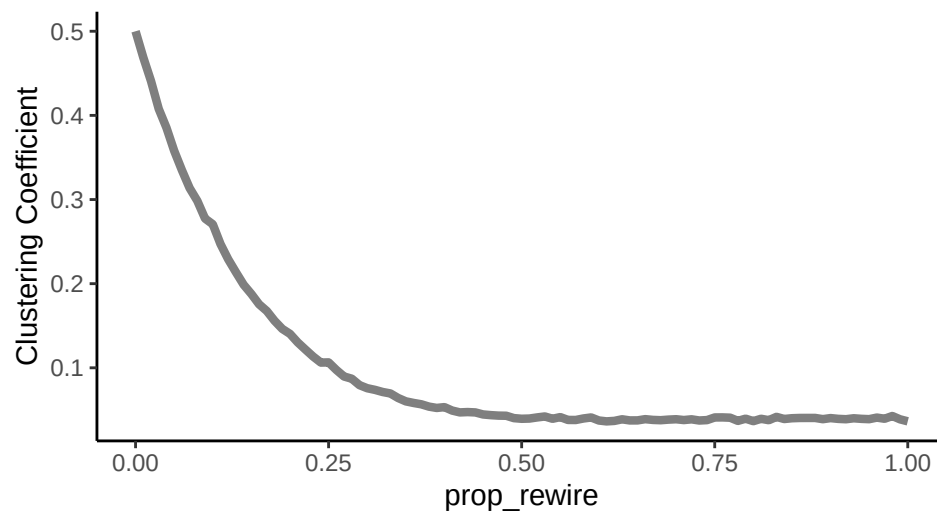


```

results.mat %>%
  as_data_frame() %>%
  group_by(prop_rewire) %>%
  summarise(CC=mean(CC)) %>%
  ggplot(aes(x=prop_rewire, y=CC)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Watts-Strogatz (size 100, 2 neighbors)") +
  ylab("Clustering Coefficient")

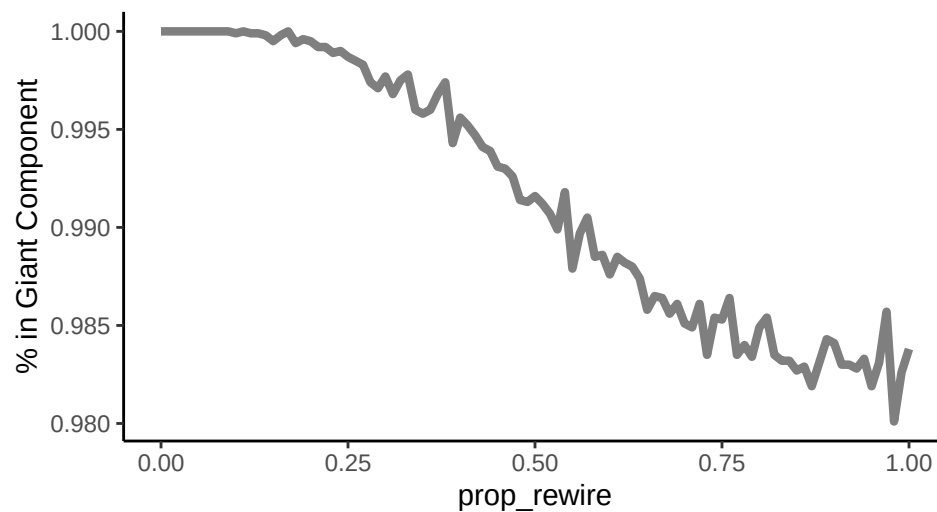
```

Watts–Strogatz (size 100, 2 neighbors)



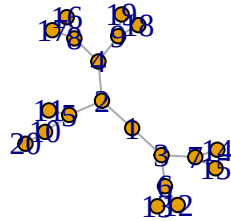
```
results.mat %>%
  as_data_frame() %>%
  group_by(prop_rewire) %>%
  summarise(p_gc=mean(p_gc)) %>%
  ggplot(aes(x=prop_rewire, y=p_gc)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Watts–Strogatz (size 100, 2 neighbors)") +
  ylab("% in Giant Component")
```

Watts–Strogatz (size 100, 2 neighbors)



Tree

```
tree20.g <- make_tree(20, children=2, mode="undirected")
plot.igraph(tree20.g)
```



```
ecount(tree20.g)
```

```
## [1] 19
```

```
vcount(tree20.g)
```

```
## [1] 20
```

```
components(tree20.g)$no
```

```
## [1] 1
```

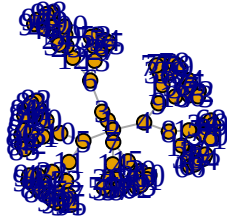
```
mean_distance(tree20.g)
```

```
## [1] 4.063158
```

```
mean(harmonic_centrality(tree20.g))
```

```
## [1] 6.455714
```

```
tree100.g <- make_tree(100, children=2, mode="undirected")
plot.igraph(tree100.g)
```



```
ecount(tree100.g)
```

```
## [1] 99
```

```
vcount(tree100.g)
```

```
## [1] 100
```

```
components(tree100.g)$no
```

```
## [1] 1
```

```
mean_distance(tree100.g)
```

```
## [1] 7.731313
```

```
mean(harmonic_centrality(tree100.g))
```

```
## [1] 16.61768
```

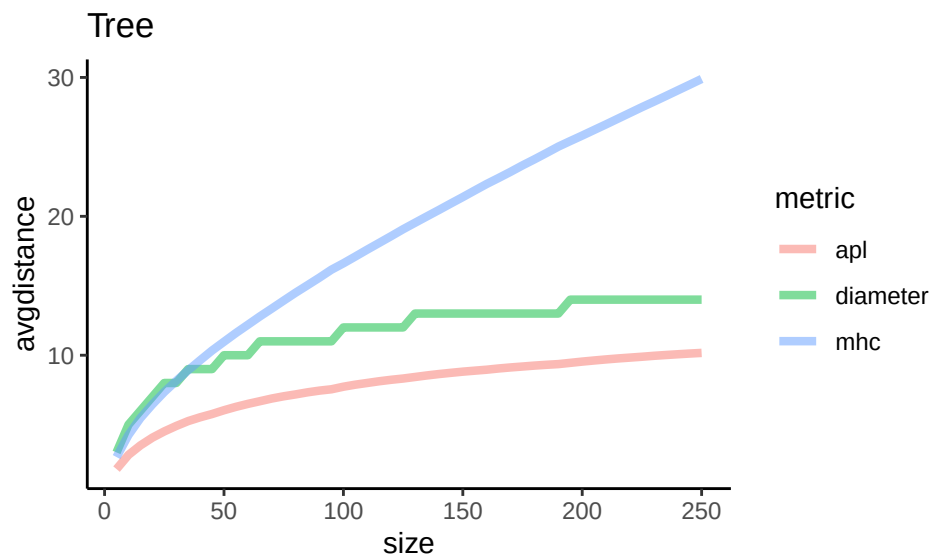
```
# you really don't need to know all this stuff with for loops and matrices  
# unless you want to do a senior thesis using simulations or the like  
trials <- 1  
testvalues <- seq(5,250,5)  
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)  
colnames(results.mat) <- c("size","apl","mhc","diameter")  
j <- 1  
for (i in testvalues) {  
  for (k in 1:trials) {  
    g <- make_tree(i, children=2, mode="undirected")
```

```

    apl <- mean_distance(g)
    mhc <- harmonic_centralty(g) %>% mean()
    diameter <- diameter(g)
    results.mat[j,] <- c(i, apl, mhc, diameter)
    j <- j+1
  }
}

results.mat %>%
  as_data_frame() %>%
  group_by(size) %>%
  summarise(apl=mean(apl),mhc=mean(mhc),diameter=mean(diameter)) %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Tree")

```

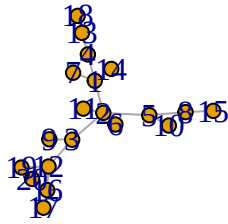


Barabasi (“preferential attachment” or “scale free”)

```

pa20.g <- sample_pa(20,directed = F)
plot.igraph(pa20.g)

```



```
ecount(pa20.g)
```

```
## [1] 19
```

```
vcount(pa20.g)
```

```
## [1] 20
```

```
components(pa20.g)$no
```

```
## [1] 1
```

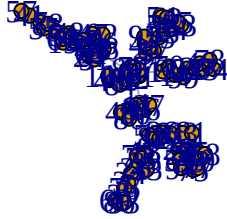
```
mean_distance(pa20.g)
```

```
## [1] 3.589474
```

```
mean(harmonic centrality(pa20.g))
```

```
## [1] 6.963214
```

```
pa100.g <- sample_pa(100,directed = F)
plot.igraph(pa100.g)
```



```
ecount(pa100.g)
```

```
## [1] 99
```

```
vcount(pa100.g)
```

```
## [1] 100
```

```
components(pa100.g)$no
```

```
## [1] 1
```

```
mean_distance(pa100.g)
```

```
## [1] 5.332727
```

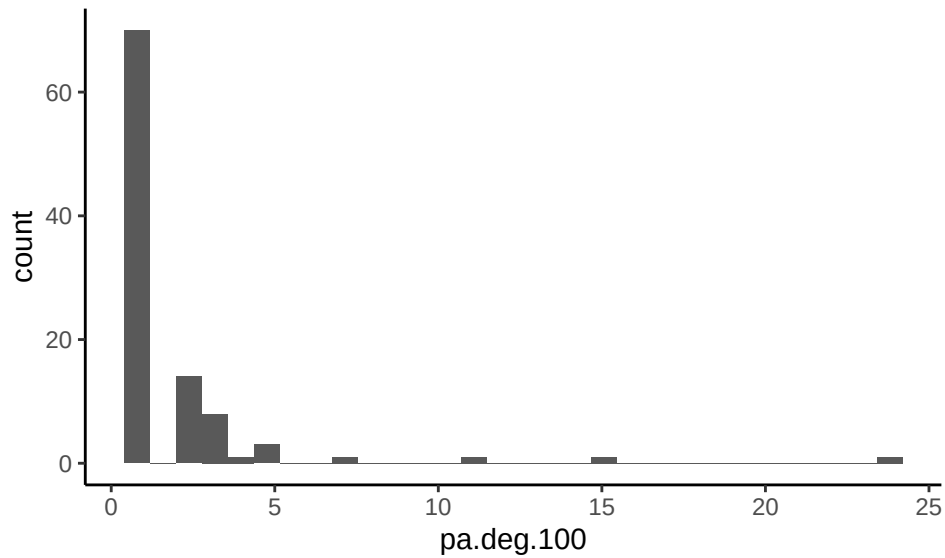
```
mean(harmonic_centrality(pa100.g))
```

```
## [1] 22.96144
```

<https://www.netlogoweb.org/launch#https://www.netlogoweb.org/assets/modelslib/Sample%20Models/Networks/Preferential%20Attachment.nlogo>

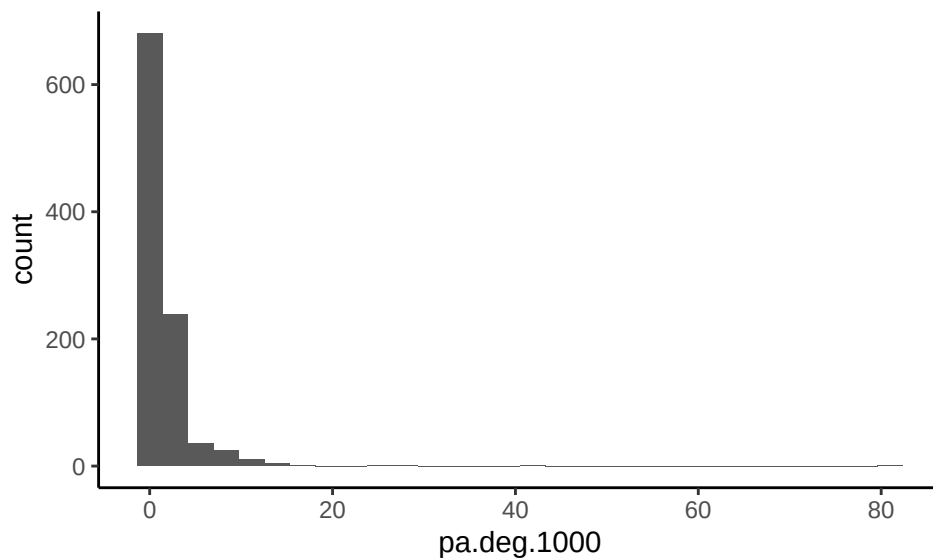
```
pa.deg.100 <- sample_pa(100) %>% degree()
ggplot(mapping = aes(pa.deg.100)) + geom_histogram() + theme_classic()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



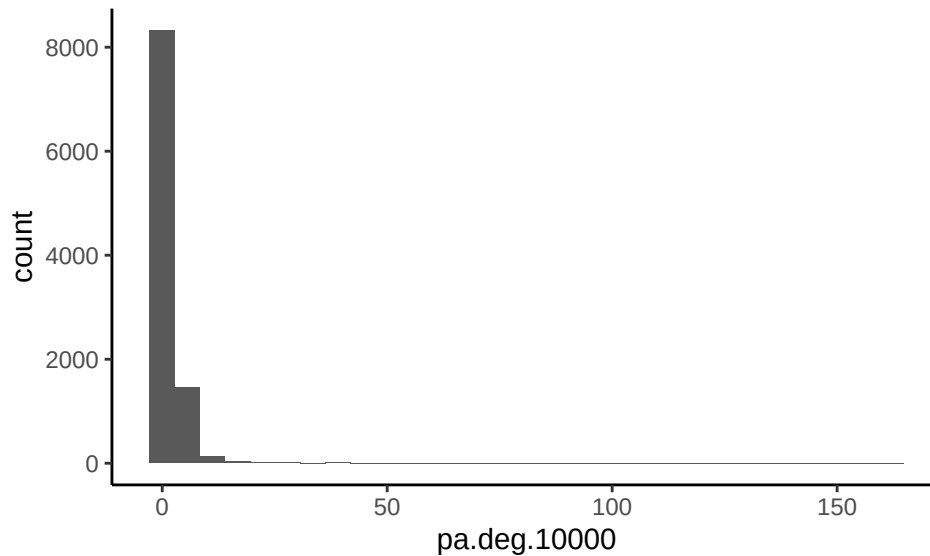
```
pa.deg.1000 <- sample_pa(1000) %>% degree()
ggplot(mapping = aes(pa.deg.1000)) + geom_histogram() +theme_classic()
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



```
pa.deg.10000 <- sample_pa(10000) %>% degree()
ggplot(mapping = aes(pa.deg.10000)) + geom_histogram() +theme_classic()
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



```
# you really don't need to know all this stuff with for loops and matrices
# unless you want to do a senior thesis using simulations or the like
trials <- 20
testvalues <- seq(5,250,5)
results.mat <- matrix(nrow = length(testvalues)*trials, ncol = 4)
colnames(results.mat) <- c("size", "apl", "mhc", "diameter")
j <- 1
for (i in testvalues) {
  for (k in 1:trials) {
    g <- sample_pa(i, directed = F)
    apl <- mean_distance(g)
    mhc <- harmonic_centralities(g) %>% mean()
    diameter <- diameter(g)
    results.mat[j,] <- c(i, apl, mhc, diameter)
    j <- j+1
  }
}

results.mat %>%
  as_data_frame() %>%
  group_by(size) %>%
  summarise(apl=mean(apl), mhc=mean(mhc), diameter=mean(diameter)) %>%
  pivot_longer(!size, names_to = "metric", values_to = "avgdistance") %>%
  ggplot(aes(x=size, y=avgdistance, colour = metric)) +
  geom_line(lwd=1.5, alpha=0.5) +
  theme_classic() +
  ggtitle("Preferential Attachment")
```

